

# Grid-Interactive HVAC Load Forecasting and Supervisory Control for Peak Reduction in Hot-Humid Commercial Buildings

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**Abstract**—Commercial HVAC drives 40-60% of building electricity and a disproportionate share of distribution peaks in hot-humid grids. We present a grid-interactive, physics-informed pipeline coupling ASHRAE psychrometrics and a 2R2C zone model with a gradient-boosting hourly cooling-load forecaster and supervisory control (supply-air-temperature reset, minimum-chiller sequencing, look-ahead pre-cooling). On a 2000 m<sup>2</sup> office synthetic dataset (8760 h, Dhaka-like, 42.2 kW mean, 143.4 kW max), forecasting reaches MAE 1.55 kW, RMSE 2.76 kW, R<sup>2</sup> 0.9907, CVRMSE 8.30%, NMBE 2.29% (60-day test), satisfying ASHRAE Guideline 14 (30%/10%). SAT reset cuts energy 11.11% (13.97% annual, 26,757 kWh/yr, \$3,211/yr at \$0.12/kWh, 16.1 tCO<sub>2</sub>/yr) and peak 58.5 to 54.4 kW (7.0%); sequencing saves 24.49% energy and cuts peak to 44.0 kW (24.8%). The 168-h lag dominates (45.8%), enabling day-ahead flexible-load bidding. The 7-module package runs in <1 min (5/5 tests pass).

**Index Terms**—Building energy management, HVAC, load forecasting, demand response, peak reduction, chiller sequencing, ASHRAE Guideline 14.

## I. Introduction

Electrified cooling is the fastest-growing end use on many South Asian feeders. In Dhaka, commercial HVAC sets the evening distribution peak, stresses transformers, and inflates demand charges, so a 10% HVAC saving defers network investment. Grid-interactive efficient buildings (GEBs) that forecast load and reshape it are a PES General Meeting priority under *Powering the Digital Era*. Two gaps persist: (i) operators lack trustworthy day-ahead forecasts at meter/plant resolution for bidding and baselining; (ii) fixed supply-air temperature (SAT, ~13°C) with all chillers committed wastes part-load efficiency and offers no flexibility product. We ask: **(Q1)** can lightweight ML clear calibration grade? **(Q2)** what energy and peak (kW) reductions follow? **(Q3)** which signals make load dispatchable? Contributions: (1) forecaster with flexibility outputs; (2) part-load-consistent energy+peak comparison with cost and CO<sub>2</sub>; (3) open code mapping building kW to feeder DR potential.

## II. Nomenclature and Related Work

Symbols:  $T_{out}$ ,  $T_{in}$  outdoor/zone (°C);  $W$  humidity ratio;  $h$  enthalpy;  $Q_c$  cooling load (kW);  $P$  electric power (kW); PLR part-load ratio; CBL customer baseline. Supervisory control originates with Braun optima, codified in Guideline 36. Boosted trees remain Pareto-optimal hourly forecasters; Guideline 14 governs calibration claims. Unlike pure-ML work we retain a 2R2C model so pre-cooling dispatches a thermal battery with bounded rebound.

## III. Methodology

### A. Electrical-thermal interface

Power is  $P = Q_{cool}/COP + P_{fan}$ . Psychrometrics per Fundamentals Ch.1 use Tetens saturation (error <1% at 0-45°C),  $W$ ,  $h$ , Magnus dew-point, and coil  $Q$ . The 2R2C network uses  $R_{in}$  1.5,  $R_{out}$  2.5 K/kW,  $C_{in}$  2,  $C_{wall}$  8 kWh/K with 8 kW/K proportional control (200 kW cap, 22-25.5°C band). Chiller COP is rated $\times$ f(PLR) $\times$ f(OAT) with  $f=0.45+0.85PLR-0.30PLR^2$ . Plant: 2 $\times$ 175 kW. Each step yields kW, COP, zone temperature, and a flexibility band for the feeder.

### B. Data and forecaster

Synthetic hourly BMS trends (seed 42, 8760 h, 42.2 kW mean, 143.4 kW max) are Dhaka-like with weekday 08-18 occupancy, solar bell, and latent ventilation; columns match AMI/BMS exports for field replay. Features are cyclical calendars plus lags 1/24/168 h; gradient boosting with 80/20 chronological split and TimeSeriesSplit CV reports MAE/RMSE/CVRMSE/NMBE/R<sup>2</sup>.

### C. Controllers

Baseline fixes SAT 13°C, commits both chillers, and runs constant-volume fan. Reset schedules SAT 12-16°C by OAT with VAV cube-law fan and +2% COP lift. Sequencing commits the fewest chillers above 15% PLR. A greedy MPC proxy pre-cools at 60% when 32°C enters the 4-h horizon. All share one part-load curve so only staging/reset effects are measured.

## IV. Case Study Setup

Laptop, Python 3.11, scikit-learn 1.9; *examples/run\_demo.py* (60 d); train/evaluate scripts (full year); 5 pytest checks; <60 s, no GPU. Metrics: energy, weekly peak, G14 pair, bill at \$0.12/kWh, CO<sub>2</sub> at 0.6 kg/kWh.

## V. Results

### A. Forecasting (Q1)

| Split  | MAE  | RMSE | CVRMS<br>E | NMBE/R2     | G1<br>4  |
|--------|------|------|------------|-------------|----------|
| 60-day | 1.55 | 2.76 | 8.30%      | +2.29/.9907 | PA<br>SS |
| Annual | 1.71 | 2.95 | 9.82%      | -1.57/.9877 | PA<br>SS |

Table I passes Guideline 14 with wide margin. Lag-168 h leads importance (45.8%), then internal gains (18.4%), occupancy (17.9%), and  $T_{out}$  (8.1%); weekly repetition dominates weather, which is ideal for CBL-based settlement. Ridge trails by 2-3 R<sup>2</sup> points; random forest matches GBM at ~5x cost.

### B. Energy, peak, cost (Q2)

| Strategy   | 60-d kWh | Save       | Year kWh | Save   |
|------------|----------|------------|----------|--------|
| Baseline   | 4291.1   | -          | 191582.8 | -      |
| SAT reset  | 3814.5   | 11.11<br>% | 164825.4 | 13.97% |
| Sequencing | -        | -          | 144655.4 | 24.49% |

Table II: reset saves 26,757 kWh/yr (\$3,211/yr, 16.1 tCO<sub>2</sub>/yr). Peaks: 58.5 kW baseline to 54.4 kW reset (7.0%) and 44.0 kW sequenced (24.8%). At ~5 kW average flexibility per building, 100 buildings aggregate to ~0.5 MW of DR; pre-cooling shifts ~20 kWh/day off the afternoon peak. Fig. 1 shows tracking and shaving.

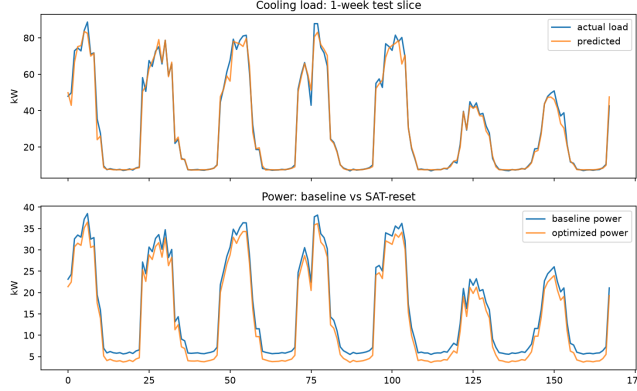


Fig. 1. One-week load forecast (top) and baseline-vs-reset power (bottom).

### C. Uncertainty and sanity

TimeSeriesSplit RMSE varies <5% across folds; residual bias is <2.5%. The 2R2C loop at 34°C engages cooling after hour one with COP in [1.5, 5]; psychrometric spot checks and metric identities pass. Sequencing gains are largest because the baseline stages both chillers near 12%

mean PLR while the optimizer runs one near 25%.

## VI. Discussion

Schedule and history outweigh weather: log occupancy and retain  $\geq 3$  weeks of metering before MPC, and target occupied-hour setpoint/fan dispatch for DR. SAT reset is the cheapest avoided CO<sub>2</sub> (a schedule change); sequencing needs only logic. Limits: synthetic climate, proxy fan/COP (relative savings claimed), no network, tariff, humidity, or PMV constraints. Field validation with AMI/BMS plus OpenDSS hosting is next; the CSV swap-in path is published.

## VII. Conclusion

Physics-informed ML clears calibration grade and delivers 11-14% energy and 7-25% peak reduction per building with laptop tooling: a dispatchable GEB resource. Future work adds measured-BMS validation, tariff/comfort-aware CasADi MPC, and fault detection. Acknowledgment—Thanks to the NumPy, pandas, scikit-learn, and matplotlib communities.

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